Matrix encoding method in variational algorithm of calculating eigenvalues and generalized eigenvalues

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We propose a variational method for constructing the eigenvalues and generalized eigenvalues for an arbitrary $N \times N$ complex matrix. The quantum part of our algorithm is based on encoding the matrix elements into the pure state of a quantum system and expressing the loss function with optimization parameters in terms of certain probability amplitudes in the superposition state. The principal step of this algorithm is the measurement of the ancilla state that removes all extra terms from the above superposition and allows to probabilistically construct the required loss function along with its derivatives with respect to the optimization parameters. These output data are used to find the new values of optimization parameters for the next iteration of the loss function in the gradient optimization method. The depth and size of the circuit for this algorithm are, respectively, $O(N^2 \log N)$ and $O(\log N)$.

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I. INTRODUCTION

During last decades, quantum informatics experiences huge progress been involved in such global problems as communication [1–10], cryptography [11–13] and computation [14–16]. The later, although been most attractive application of quantum physics, faces set of series obstacles for development due to requirement of organizing the quantum platform with large number of qubits admitting reliable pairwise interaction between any two of them. Nevertheless, the theoretical background of quantum computation is rather well developed [17, 18], and there is significant progress in quantum architecture [19, 20]. Intensive research in this direction is stimulated by predicted exponential speedup of quantum computation algorithms which provides doubtless advantage of these algorithms over their classical counterparts.

Overlooking achievements in development of quantum algorithms, first of all we refer to Deutsch algorithm [21] exploring quantum parallelism, Grover search algorithm [22], Shor's factorization algorithm [23, 24]. Extensive applicability is deserved by quantum Fourier transform [17, 25, 26] and phase estimation algorithm [17, 27].

However, at present time, not any classical algorithm has its quantum counterpart. In particular, the quantum optimization algorithms, such as GRadient Ascent Pulse Engineering (GRAPE) approach [28] and Chopped Random Basis (CRAB) optimization method [29], whose application to control problems in various physical areas is wildly acknowledged [30–36], have not been constructed yet because of the principal obstacles for realizing the computational cycles in quantum algorithms as well as the real-time re-encoding the optimization parameters of unitary rotations at each iteration during algorithm evaluation. This is the reason that gives rise to intensive development of so-called hybrid algorithms that combines both classical and quantum subroutines.

Among them, we select the family of variational quantum algorithms which include the classical optimization algorithm reaching the optimal values of the optimization parameters using the loss (or objective) function calculated by means of quantum algorithm. Thus, the variational quantum eigensolver allowing to estimate the ground state energy of Hamiltonians was proposed in Ref.[37]. The variational algorithms for finding eigenvalues and generalized eigenvalues by scanning the energy interval were proposed, respectively, in Refs. [38] and [39]. The algorithm for constructing the minimal generalized eigenvalue for Hermitian matrices was developed in Ref. [40]. In [41], all generalized eigenvalues of Hermitian operators where found successively using the Euclidean time evolution. The significant promotion was realized in [42], where the algorithm for calculating eigenvalues by reducing the matrix to the triangular form was developed for an arbitrary matrix. The gradient optimization method with effective quantum calculation of loss function together with its derivatives with respect to optimization parameters was implemented in variational quantum algorithm for contracting the singular value decomposition [43] (this algorithm was verified by its simulating on the PaddlePaddle Deep Learning Platform [44–46]) and the generalized eigenvalues [47]. We note that the nonunitarity obstacle for considering arbitrary matrices in the above quoted variational algorithms was solved by using the linear combination of unitaries (LCU) [42, 43], by transforming the matrix to the Hermitian form in the case of evolutionary approach [38], or, alternatively, by the matrix encoding using specially reordered matrix elements [48]. The interest in development of algorithms for solving generalized eigenvalue problem is supported by its wide application not only in physics, statistic and mathematics, but also in new research areas such as machine learning [49, 50], quantum chemistry [51] and algo rithms operating with large computational data [52, 53].

In our paper, we propose the variation algorithm for calculating generalized eigenvalues for arbitrary complex matrices A and B using transformation of these matrices to the triangular form [47]. The novelty is encoding these matrices into the quantum state of certain quantum system [54, 55], which, in particular, allows to avoid the problem of non-unitarity of considered matrices. Thus, the matrices A and B are considered as a quantum state of a particular quantum system rather than in terms of unitary operators as in the most of traditional variational approaches to the eigenvalue problem cited above. We also apply the similar approach of matrix triangularization and encoding to the variational algorithm for calculating the usual eigenvalues of A and concentrate on some specific features of this algorithm. Remark that the method of matrix encoding was first applied to the matrix manipulation algorithms (addition, multiplication, inversion, linear system solver) in [54, 56]. The principal novelty and advantage of this method is that it uses encoding the matrix elements into the probability amplitudes of the superposition state of certain quantum system and does not require embedding the input matrices into the Hermitian operators with subsequent approximate exponentiation via Suzuki-Trotter [57–60] or Baker-Champbell-Hausdorff [61] methods, although the theoretical background for applying the latter methods in matrix algebra is rather well developed [14, 15, 62–65]. Later, the matrix-encoding method was applied to the variational algorithm for constructing singular value decomposition of an arbitrary complex matrix [55].

In the present work, we combine the quantum algorithm for calculating the loss function (together with its derivatives with respect to the optimization parameters) at fixed values of optimization parameters with classical gradient algorithm of calculating the values of the optimization parameters for the next iteration step. Finally, we study the relation among the accuracies for calculating the loss function, eigenvalues and optimization parameters for both algorithms calculating eigenvalues and generalized eigenvalues. All theoretical arguments and statement are justified by numerical simulations of both variational algorithms for 4×4 randomly generated complex matrices.

The paper is organized as follows. The variational algorithm for calculating generalized eigenvalues (GEV) based on matrix encoding into the pure quantum state is proposed in Sec. II. Features of the algorithm constructing usual eigenvalues (EV) are discussed in Sec.III. Characteristics of both algorithms including circuit parameters and parameters of the optimization algorithm together with accuracy of calculation are considered in Sec.IV. Numerical constructing of GEV and EV for 4×4 matrices and analysis of accuracy of the calculated eigenvalues and accuracy of encoding the optimization parameters are presented in Sec.V. Obtained result are discussed in Sec.VI.

II. VARIATIONAL ALGORITHM FOR CONSTRUCTING GENERALIZED EIGENVALUES

A. Preliminaries

We consider the variational algorithm for constructing GEV for two arbitrary non-degenerated complex $N \times N$ matrices A and B assuming $N = 2^n$,

$$A|\psi\rangle = \lambda B|\psi\rangle, \tag{1}$$

where λ is the eigenvalue corresponding to the eigenvector $|\psi\rangle$. According to the generalized Schur decomposition theory [66], for any matrices $A, B \in \mathbb{C}^{N\times N}$, there exist the unitary matrices Q and Z such that

$$Q^\dagger AZ = T, \;\; Q^\dagger BZ = S, \tag{2}$$

where $T = \{t_{ij}\}, S = \{s_{ij}\} \in \mathbb{C}^{N\times N}$ are upper triangular matrices. If, for some k, $t_{kk} = s_{kk} = 0$, then the appropriate λ_k can be any complex number, otherwise

$$\lambda_k = \frac{t_{kk}}{s_{kk}}, \;\; s_{kk} \neq 0, \;\; k = 0, \ldots, N-1. \tag{3}$$

We define the loss function as the sum of squares of absolute values of all vanishing elements in T and S [47]:

$$L(\alpha,\beta) = \sum_{j=1}^{N-1}\sum_{j=0}^{i-1}\left(|\langle i|U^T(\alpha)AU(\beta)|j\rangle|^2 + |\langle i|U^T(\alpha)BU(\beta)|j\rangle|^2\right) = \tag{4}$$
$$\sum_{j=1}^{N-1}\sum_{j=0}^{i-1}\left(|\langle i|T(\alpha,\beta)|j\rangle|^2 + |\langle i|S(\alpha,\beta)|j\rangle|^2\right),$$

where the superscript T denotes matrix transposition, $\alpha = \{\alpha_0, \ldots, \alpha_{nM}\}$ and $\beta = \{\beta_0, \ldots, \beta_{nM}\}$ represent two sets of real optimization parameters, M is some integer associated with the subroutine used for preparing the unitary transformation U in Sec.II B 1. In Eq. (4), U is an arbitrary $N \times N$ unitary transformation of general form, therefore $U(\alpha)$ and $U(\beta)$ can be regarded as two arbitrary independent transformations if only α and β are two independent sets of parameters. We appeal to the gradient method of finding such parameters α_0 and β_0 that minimize the loss function, i.e.,

$$\min_{\alpha,\beta} L(\alpha,\beta) = L(\alpha_0,\beta_0). \tag{5}$$

It is shown in [47] that the loss function (4) reaches its global minimum of zero if and only if $T(\alpha,\beta)$ and $S(\alpha,\beta)$ are upper triangular matrices. Then

$$\begin{aligned} &T = U^T(\alpha_0)AU(\beta_0), \quad S = U^T(\alpha_0)BU(\beta_0), \\ &Q^\dagger = U^T(\alpha_0), \;\; Z = U(\beta_0), \end{aligned} \tag{6}$$

and the generalized eigenvalues can be found using Eqs.(3).

B. Quantum algorithm calculating loss function

To construct the variational quantum algorithm for calculating generalized eigenvalues, we introduce six n-qubit subsystems: the subsystems R and C serve to enumerate, respectively, the rows and column of A and B, the subsystems χ, $\tilde{\chi}$ and ψ, $\tilde{\psi}$ are needed to operate with $\langle j|U^T(\alpha)$ and $U(\beta)|j\rangle$ in (4). In addition, the one-qubit subsystem L serves to mark the elements of matrices A and B in the superposition state (see Eq.(8)), and the one-qubit subsystem K serves as a controlling qubit in succeeding controlled operations. At the last steps of the algorithm we will introduce two one-qubit ancilae $\tilde{B}$ and B to properly organize garbage removal via measurement.

First of all, we have to prepare the above mentioned matrices $A = \{a_{ij} : i,j = 0,\dots,N-1\}$ and $B = \{b_{ij} : i,j = 0,\dots,N-1\}$ for encoding into the state of a quantum system. To this end, we normalize these matrices assuming that $|a_{00}|^2 + |b_{00}|^2 \neq 0$, i.e., replace A and B with

$$\tilde{A} = \frac{A}{\sqrt{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(|a_{ij}|^2 + |b_{ij}|^2)}}, \quad \tilde{B} = \frac{B}{\sqrt{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(|a_{ij}|^2 + |b_{ij}|^2)}}. \tag{7}$$

Hereafter we do not write tilde over A and B. Now we can encode the elements of the matrices A and B into the superposition state of R, C and L using, for instance, the state-encoding algorithm developed in Ref.[67] and obtain the following superposition state:

$$|\Psi\rangle_{RCL} = |A\rangle + |B\rangle = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(a_{ij}|0\rangle_L + b_{ij}|1\rangle_L)|i\rangle_R|j\rangle_C, \tag{8}$$
$$\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(|a_{ij}|^2 + |b_{ij}|^2) = 1,$$

where the normalization is provided by Eqs.(7). In other words, we have quantum access to the matrices A and B [68]. Subsystems χ, $\tilde{\chi}$, ψ, $\tilde{\psi}$ and K are in the ground states initially, i.e., the initial state of the whole system reads:

$$|\Phi_0\rangle = |\Psi\rangle_{RCL}|0\rangle_\chi|0\rangle_{\tilde{\chi}}|0\rangle_\psi|0\rangle_{\tilde{\psi}}|0\rangle_K. \tag{9}$$

Hereafter in this paper, the subscript at the operator means the subsystem to which this operator is applied.

The quantum algorithm is illustrated by the circuit in Fig.1a. As the first step, we apply the Hadamard transformation to each qubit of χ, ψ and K (we denote this set of transformations by $W^{(0)}_{\chi\psi K} = H_\chi H_\psi H_K$):

$$|\Phi_1\rangle = W^{(0)}_{\chi\psi K}|\Phi_0\rangle = \frac{1}{2^{(2n+1)/2}}\sum_{k=0}^{N-1}\sum_{l=0}^{N-1}|\Psi\rangle_{RCL}|k\rangle_\chi|0\rangle_{\tilde{\chi}}|l\rangle_\psi|0\rangle_{\tilde{\psi}}(|0\rangle_K + |1\rangle_K), \tag{10}$$

creating two systems of orthonormal states $|k\rangle_\chi$ and $|l\rangle_\psi$, $k,l = 0,\dots,N-1$, together with the superposition state of the controlling qubit K.

Now we double those states $|k\rangle_\chi$ and $|l\rangle_\psi$ that are linked to the excited state of the one-qubit subsystem K and create the same states $|k\rangle_{\tilde{\chi}}$ and $|l\rangle_{\tilde{\psi}}$ of the subsystems $\tilde{\chi}$ and $\tilde{\psi}$. For this purpose we introduce the projectors

$$P_{\chi_i K} = |1\rangle_{\chi_i}|1\rangle_K\, {}_{\chi_i}\langle 1|_K\langle 1|, \quad P_{\psi_i K} = |1\rangle_{\psi_i}|1\rangle_K\, {}_{\psi_i}\langle 1|_K\langle 1|, \quad i = 1,\dots,n, \tag{11}$$

and the controlled operator

$$W^{(1)}_{\chi\tilde{\chi}\psi\tilde{\psi}K} = \prod_{i=1}^{n}\Big(P_{\chi_i K}\otimes\sigma^{(x)}_{\tilde{\chi}_i} + (I_{\chi_i K} - P_{\chi_i K})\otimes I_{\tilde{\chi}_i}\Big)\Big(P_{\psi_i K}\otimes\sigma^{(x)}_{\tilde{\psi}_i} + (I_{\psi_i K} - P_{\psi_i K})\otimes I_{\tilde{\psi}_i}\Big), \tag{12}$$

whose depth is $O(n)$. Hereafter the subscript attached to the notation of a subsystem indicates the appropriate qubit of this subsystem. Thus, the subscript i in the right hand side of Eq.(12) denotes the ith qubit of the appropriate subsystem χ, $\tilde{\chi}$, ψ, $\tilde{\psi}$. Applying $W^{(1)}_{\chi\tilde{\chi}\psi\tilde{\psi}K}$ to $|\Phi_1\rangle$ we obtain

$$|\Phi_2\rangle = W^{(1)}_{\chi\tilde{\chi}\psi\tilde{\psi}K}|\Phi_1\rangle = \frac{1}{2^{(2n+1)/2}}\left(\sum_{k,l=0}^{N-1}|\Psi\rangle_{RCL}|k\rangle_\chi|0\rangle_{\tilde{\chi}}|l\rangle_\psi|0\rangle_{\tilde{\psi}}|0\rangle_K \right. \tag{13}$$
$$\left. + \sum_{k,l=0}^{N-1}|\Psi\rangle_{RCL}|k\rangle_\chi|k\rangle_{\tilde{\chi}}|l\rangle_\psi|l\rangle_{\tilde{\psi}}|1\rangle_K)\right) + |g_2\rangle,$$

where the garbage $|g_2\rangle$ collects all extra terms, it is orthogonal to the first term in Eq.(13).

Next, we prepare and apply the unitary operators $U_\chi(\alpha)$ and $U_\psi(\beta)$ (see Eq. (4)) controlled by the excited state of the subsystem K by means of the operator $W^{(2)}_{\chi\psi K}$:

$$W^{(2)}_{\chi\psi K} = W^{(2)}_{\chi K}W^{(2)}_{\psi K}, \tag{14}$$
$$W^{(2)}_{\chi K} = |1\rangle_K\,{}_K\langle 1|\otimes U_\chi(\alpha) + |0\rangle_K\,{}_K\langle 0|\otimes I_\chi, \quad W^{(2)}_{\psi K} = |1\rangle_K\,{}_K\langle 1|\otimes U_\psi(\beta) + |0\rangle_K\,{}_K\langle 0|\otimes I_\psi. \tag{15}$$

It is convenient to represent the action of the operators U_χ and U_ψ on the vectors $|k\rangle_\chi$ and $|k\rangle_\psi$ in terms of their matrix elements as follows:

$$U_\chi(\alpha)|k\rangle_\chi = \sum_{l_1=0}^{N-1} u_{l_1 k}(\alpha)|l_1\rangle_\chi,\;\; U_\psi(\beta)|l\rangle_\psi = \sum_{l_2=0}^{N-1} u_{l_2 l}(\beta)|l_2\rangle_\psi. \tag{16}$$

The depth of the operator $W^{(2)}_{\chi\psi K}$ will be discussed in Sec.II B 1. Then, applying $W^{(2)}_{\chi\psi K}$ to $|\Phi_2\rangle$ we obtain

$$\begin{aligned}|\Phi_3\rangle &= W^{(2)}_{\chi\psi K}|\Phi_2\rangle \\ &= \frac{1}{2^{(2n+1)/2}}\left(\sum_{k,l=0}^{N-1}|\Psi\rangle_{RCL}|k\rangle_\chi|0\rangle_{\tilde\chi}|l\rangle_\psi|0\rangle_{\tilde\psi}|0\rangle_K\right.\\ &+\left.\sum_{k,l=0}^{N-1}\sum_{l_1=0}^{N-1}\sum_{l_2=0}^{N-1}u_{l_1k}(\alpha)u_{l_2l}(\beta)|\Psi\rangle_{RCL}|l_1\rangle_\chi|k\rangle_{\tilde\chi}|l_2\rangle_\psi|l\rangle_{\tilde\psi}|1\rangle_K)\right)+|g_3\rangle.\end{aligned} \tag{17}$$

Here, the garbage $|g_3\rangle$ collects all states that must be removed, it is orthogonal to the first term in (17). To multiply the matrices A and B by the matrices $U^T_\chi(\alpha)\equiv U^T(\alpha)$ (from the left) and $U_\psi(\beta)\equiv U(\beta)$ (from the right) and eventually calculate the sums $\sum_{i,j}\,|_\psi\langle i|U^T(\alpha)AU(\beta)|j\rangle_\psi|^2$, $\sum_{i,j}\,|_\psi\langle i|U^T(\alpha)BU(\beta)|j\rangle_\psi|^2$ in (4), we proceed according to Refs.[54, 56]. Using projectors P_{χ_iK} and P_{ψ_iK}, given in Eqs.(11), we introduce the following controlled operators:

$$\begin{aligned}C^{(1)}_{\chi KR} &= \prod_{i=1}^{n}\Big(P_{\chi_iK}\otimes\sigma^{(x)}_{R_i}+(I_{\chi_iK}-P_{\chi_iK})\otimes I_{R_i}\Big),\\ C^{(2)}_{\psi KC} &= \prod_{i=1}^{n}\Big(P_{\psi_iK}\otimes\sigma^{(x)}_{C_i}+(I_{\psi_iK}-P_{\psi_iK})\otimes I_{C_i}\Big),\end{aligned} \tag{18}$$

Here, the operator $C^{(1)}_{\chi KR}$ is required for multiplying $U^T(\alpha)$ and A (B), the operator $C^{(2)}_{\psi KC}$ serves for multiplying A (B) and $U(\beta)$. Applying the operator $W^{(3)}_{RC\chi\psi K}=C^{(2)}_{\psi KC}C^{(1)}_{\chi KR}$, whose depth is $O(n)$, to the state $|\Phi_3\rangle$ we obtain

$$\begin{aligned}&|\Phi_4\rangle = W^{(3)}_{RC\chi\psi K}|\Phi_3\rangle\\ &=\frac{1}{2^{(2n+1)/2}}\left(\sum_{k,l=0}^{N-1}\big(a_{00}|0\rangle_L+b_{00}|1\rangle_L\big)|0\rangle_R|0\rangle_C|k\rangle_\chi|0\rangle_{\tilde\chi}|l\rangle_\psi|0\rangle_{\tilde\psi}|0\rangle_K\right.\\ &+\left.\sum_{k,l=0}^{N-1}\sum_{l_1=0}^{N-1}\sum_{l_2=0}^{N-1}u_{l_1k}(\alpha)u_{l_2l}(\beta)\big(a_{l_1l_2}|0\rangle_L+b_{l_1l_2}|1\rangle_L\big)|0\rangle_R|0\rangle_C|\rangle|l_1\rangle_\chi|k\rangle_{\tilde\chi}|l_2\rangle_\psi|l\rangle_{\tilde\psi}|1\rangle_K\right)+|g_4\rangle\end{aligned} \tag{19}$$

where the first part in the right hand side collects the terms needed for further calculations (these terms will be labelled later on by the operator $W^{(6)}_{RC\chi\psi\tilde BB}$, see eq.(28)) and $|g_4\rangle$ (orthogonal to the first term in (19)) is the garbage to be removed. Now, to complete the multiplications $U^T(\alpha)AU(\beta)$ and $U^T(\alpha)BU(\beta)$ according to the multiplication algorithm (see Appendix in Ref.[56]), we introduce the operator

$$W^{(4)}_{\chi\psi} = H_\chi H_\psi, \tag{20}$$

where H_χ and H_ψ are the sets of Hadamard transformations applied to each qubit of the subsystems χ and ψ respectively, and apply $W^{(4)}_{\chi\tilde\chi\psi}$ to the state $|\Phi_4\rangle$. Selecting only the needed terms and moving others to the garbage $|g_5\rangle$, we obtain

$$\begin{aligned}&|\Phi_5\rangle = W^{(4)}_{\chi\tilde\chi\psi}|\Phi_4\rangle\\ &=\frac{1}{2^{(4n+1)/2}}\left(\big(\tilde a_{00}|0\rangle_L+\tilde b_{00}|1\rangle_L\big)|0\rangle_{\tilde\chi}|0\rangle_{\tilde\psi}|0\rangle_K+\sum_{k,l=0}^{N-1}\big(A_{kl}|0\rangle_L+B_{kl}|1\rangle_L\big)|k\rangle_{\tilde\chi}|l\rangle_{\tilde\psi}|1\rangle_K\right)|0\rangle_R|0\rangle_C|0\rangle_\chi|0\rangle_\psi+|g_5\rangle,\end{aligned} \tag{21}$$

where,

$$
\begin{aligned}
&\tilde a_{00} = 2^{2n} a_{00}, \ \ \tilde b_{00} = 2^{2n} b_{00}, && (22)\\
&A_{kl}(\alpha,\beta) = \sum_{l_1=0}^{N-1}\sum_{l_2=0}^{N-1} u_{l_1k}(\alpha)u_{l_2l}(\beta)\, a_{l_1l_2} = (U^T(\alpha)AU(\beta))_{kl}, && (23)\\
&B_{kl}(\alpha,\beta) = \sum_{l_1=0}^{N-1}\sum_{l_2=0}^{N-1} u_{l_1k}(\alpha)u_{l_2l}(\beta)\, b_{l_1l_2} = (U^T(\alpha)BU(\beta))_{kl}. && (24)
\end{aligned}
$$

We note that the factor 2^{2n} in the expressions for $\tilde a_{00}$ and $\tilde b_{00}$ in (22) appears because of the sum over k and l in the first term in the parenthesis of (19) which includes N^2 terms.

Now we label those terms in the sum in Eq.(21) that are needed for loss function according to the sum in Eq.(4). To this end, we introduce the projectors

$$
p_{jk} = |j\rangle_{\tilde\chi}|k\rangle_{\tilde\psi}|1\rangle_K \,{}_{\tilde\chi}\langle j|_{\tilde\psi}\langle k|_K\langle 1|, \ \ j=1,\dots,N-1, \ \ k=0,\dots,j-1, \tag{25}
$$

the one-qubit ancilla $\tilde B$ in the ground state and the controlled operator

$$
\begin{aligned}
&W^{(5)}_{\tilde\chi\tilde\psi K\tilde B} = P_{\tilde\chi\tilde\psi K}\otimes\sigma^{(x)}_{\tilde B} + (I_{\tilde\chi\tilde\psi K} - P_{\tilde\chi\tilde\psi K})\otimes I_{\tilde B},\\
&P_{\tilde\chi\tilde\psi K} = \sum_{j=1}^{N-1}\sum_{k=0}^{j-1} p_{jk},
\end{aligned} \tag{26}
$$

whose depth is $O(N^2\log N)$. Applying $W^{(5)}_{\tilde\chi\tilde\psi K\tilde B}$ to $|\Phi_5\rangle|0\rangle_{\tilde B}$ we obtain

$$
\begin{aligned}
&|\Phi_6\rangle = W^{(5)}_{\tilde\chi\tilde\psi K\tilde B}|\Phi_5\rangle|0\rangle_{\tilde B}\\
&= \frac{1}{2^{(4n+1)/2}}\left(\big(\tilde a_{00}|0\rangle_L + \tilde b_{00}|1\rangle_L\big)|0\rangle_{\tilde\chi}|0\rangle_{\tilde\psi}|0\rangle_K + \sum_{k=1}^{N-1}\sum_{l=0}^{k-1}\big(A_{kl}|0\rangle_L + B_{kl}|1\rangle_L\big)|k\rangle_{\tilde\chi}|l\rangle_{\tilde\psi}|1\rangle_K\right)\times\\
&|0\rangle_R|0\rangle_C|0\rangle_\chi|0\rangle_\psi|1\rangle_{\tilde B} + |g_6\rangle,
\end{aligned} \tag{27}
$$

Now we label and remove the garbage $|g_5\rangle$ from the state (27) via the measurement. To this end, we introduce another one-qubit ancillae B in the ground state and the controlled operator

$$
\begin{aligned}
&W^{(6)}_{RC\chi\psi\tilde BB} = P\otimes\sigma^{(x)}_B + (I_{RC\chi\psi\tilde B} - P)\otimes I_B,\\
&P = |0\rangle_R|0\rangle_C|0\rangle_\chi|0\rangle_\psi|1\rangle_{\tilde B}\,{}_R\langle 0|_C\langle 0|_\chi\langle 0|_\psi\langle 0|_{\tilde B}\langle 1|,
\end{aligned} \tag{28}
$$

whose depth is $O(\log N)$. Then, applying $W^{(6)}_{RC\chi\psi\tilde BB}$ to the state $|\Phi_6\rangle|0\rangle_B$ we obtain

$$
\begin{aligned}
&|\Phi_7\rangle = W^{(6)}_{RC\chi\psi\tilde BB}|\Phi_6\rangle|0\rangle_B =\\
&\frac{1}{2^{(4n+1)/2}}\left(\big(\tilde a_{00}|0\rangle_L + \tilde b_{00}|1\rangle_L\big)|0\rangle_{\tilde\chi}|0\rangle_{\tilde\psi}|0\rangle_K + \sum_{k=1}^{N-1}\sum_{l=0}^{k-1}\big(A_{kl}|0\rangle_L + B_{kl}|1\rangle_L\big)|k\rangle_{\tilde\chi}|l\rangle_{\tilde\psi}|1\rangle_K\right)\times\\
&|0\rangle_R|0\rangle_C|0\rangle_\chi|0\rangle_\psi|1\rangle_{\tilde B}|1\rangle_B + |g_6\rangle|0\rangle_B.
\end{aligned} \tag{29}
$$

Now we can remove the garbage by applying the measurement operator M_B to the ancilla B. By measurement operator M_B we mean the operator that, been applied to the superposition state $\alpha|0\rangle_B + \beta|1\rangle_B$, $|\alpha|^2+|\beta|^2=1$, yields either $|0\rangle_B$ or $|1\rangle_B$ with probability, respectively, $|\alpha|^2$ or $|\beta|^2$. Thus, applying M_B to B with the desired output $|1\rangle_B$, whose probability (success probability) is

$$
p = \frac{G^2}{2^{4n+1}}, \ \ G = \sqrt{|\tilde a_{00}|^2 + |\tilde b_{00}|^2 + \sum_{k=1}^{N-1}\sum_{l=0}^{k-1}(|A_{kl}|^2 + |B_{kl}|^2)} = \sqrt{|\tilde a_{00}|^2 + |\tilde b_{00}|^2 + L}, \tag{30}
$$

we reduce the state of the remaining system to the following one:

$$|\Phi_8\rangle = |\Psi_{out}\rangle |0\rangle_R |0\rangle_C |0\rangle_\chi |0\rangle_\psi |1\rangle_{\tilde{B}} |1\rangle_B, \tag{31}$$
$$|\Psi_{out}\rangle = G^{-1}\left(\big(\tilde{a}_{00}|0\rangle_L + \tilde{b}_{00}|1\rangle_L\big)|0\rangle_{\tilde{\chi}}|0\rangle_{\tilde{\psi}}|0\rangle_K + \sum_{k=1}^{N-1}\sum_{l=0}^{k-1}\big(A_{kl}|0\rangle_L + B_{kl}|1\rangle_L\big)|k\rangle_{\tilde{\chi}}|l\rangle_{\tilde{\psi}}|1\rangle_K \right).$$

Now, the loss function L and normalization G can be probabilistically found by measuring the state of K. The probability of obtaining $|0\rangle_K$ and $|1\rangle_K$ equal, respectively

$$p_0 = \frac{|\tilde{a}_{00}|^2 + |\tilde{b}_{00}|^2}{G^2}, \quad p_1 = \frac{L}{G^2}, \quad p_0 + p_1 = 1. \tag{32}$$

Thus, we have

$$L = \big(|\tilde{a}_{00}|^2 + |\tilde{b}_{00}|^2\big)\frac{p_1}{1-p_1}, \quad G^2 = \frac{|\tilde{a}_{00}|^2 + |\tilde{b}_{00}|^2}{1-p_1}. \tag{33}$$

This is the last step of the quantum part of the hybrid algorithm for calculating GEV. Now we give details regarding the representation of the unitary operators $U_\chi(\alpha)$ and $U_\psi(\beta)$, given in Eqs.(16), in terms of one-qubit rotations and C-NOTs.

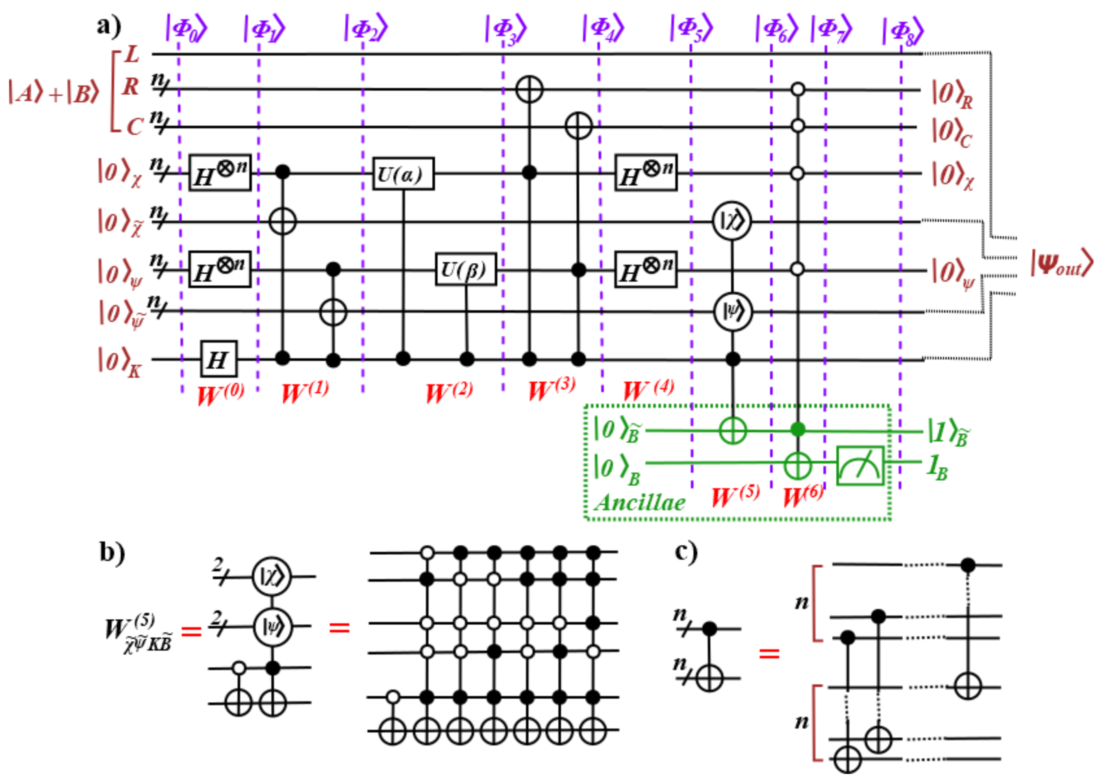


FIG. 1: The quantum circuit for calculating the loss function. Depending on M, the depth of this circuit can be estimated by either $O(M \log N)$ or $O(N^2 \log N)$ which is defined by the operator either $W^{(2)}_{\chi\psi K}$ or $W^{(5)}_{\tilde{\chi}\tilde{\psi}K\tilde{B}}$. (a) The circuit for creating the state $|\Psi_{out}\rangle$ given in (31); the operators $W^{(j)}$, $j = 0, \dots, 6$ are presented without subscripts for brevity. (b) The circuit for the operator $W^{(5)}_{\tilde{\chi}\tilde{\psi}K\tilde{B}}$ for the case of 4×4 matrices, $n = 2$. (c) The notation for the multiqubit C-not.

1. *Realization of operator* $W^{(2)}_{\chi\psi K}$

Constructing the unitary operations U_χ and U_ψ included into the operator $W^{(2)}_{\chi\psi K}$ in Eq. (14) we follow Refs. [43, 55]. The operator $W^{(2)}_{\chi\psi K}$ is the superposition of two operators $W^{(2)}_{\chi K}$ and $W^{(2)}_{\psi K}$ which are completely equivalent to each other and defer only by the parameters encoded into them. Therefore, we describe only $W^{(2)}_{\chi K}$. To realize the operator U_χ for triangularization of the complex matrices A and B, it is enough to use the one-qubit operators $r_{\chi_j}(\alpha_l) = R_z(\alpha_l^{(1)})R_y(\alpha_l^{(2)})R_z(\alpha_l^{(3)})$, where $R_z(\alpha_l^{(k)}) = \exp(-i\sigma^{(z)}\alpha_l^{(k)}/2)$, $k=1,3$, $R_y(\alpha_l^{(2)}) = \exp(-i\sigma^{(y)}\alpha_l^{(2)}/2)$ ($\sigma^{(y)}$ and $\sigma^{(z)}$ are the Pauli matrices), and C-nots [43]:

$$\begin{aligned}
&U_\chi(\alpha) = \prod_{k=1}^{M} R_k(\alpha_{(k-1)n+1},\dots,\alpha_{kn}),\quad \alpha_j = \{\alpha_j^{(1)},\alpha_j^{(2)},\alpha_j^{(3)}\},\quad \alpha=\{\alpha_j : j=1,\dots,Mn\}, \\
&R_k(\alpha_{(k-1)n+1},\dots,\alpha_{kn}) = \prod_{m=1}^{n-1} C_{\chi_m\chi_{m+1}} \prod_{j=1}^{n} r_{\chi_j}(\alpha_{(k-1)n+j}), \\
&C_{\chi_m\chi_{m+1}} = |1\rangle_{\chi_m}\,{}_{\chi_m}\langle 1| \otimes \sigma^{(x)}_{\chi_{m+1}} + |0\rangle_{\chi_m}\,{}_{\chi_m}\langle 0| \otimes I_{\chi_{m+1}}.
\end{aligned} \tag{34}$$

In this formula, R_k represents a single block of transformations encoding $3\log N$ parameters $\alpha_j^{(k)}$, $k=1,2,3$, $j=1,\dots,\log N$. Involving M blocks R_k, $k=1,\dots,M$, we enlarge the number of parameters to $3M\log N$. We note that the number of free real parameters in the $N\times N$ unitary transformation is N^2. Therefore, $M \ge \frac{N^2}{3\log N}$ to reproduce the most general unitary transformation. The realization of unitary transformation in terms of rotation operators is chosen for two reasons: (i) simple realization of U in terms of one- and two-qubit operations and (ii) simple calculation of derivatives of the loss function with respect to the optimization parameters, see Eq.(37).

Now we turn to realization of the controlled operator $W^{(2)}_{\chi K}$ given in (15) . To this end we use U_χ determined in (34) in the controlled operator $W^{(2)}_{\chi K}$, given in Eq.(15) in terms of the operators C-not, and rewrite it in the form

$$\begin{aligned}
W^{(2)}_{\chi K} = &\prod_{k=1}^{M}\prod_{m=1}^{n-1} C_{K\chi_m\chi_{m+1}} \prod_{j=1}^{n} R_{z\chi_j}(\alpha^{(1)}_{(k-1)n+j}/2) C^{(x)}_{K\chi_j} R_{z\chi_j}(\alpha^{(1)}_{(k-1)n+j}/2)\times \\
&R_{y\chi_j}(\alpha^{(2)}_{(k-1)n+j}/2) C^{(x)}_{K\chi_j} R_{y\chi_j}(\alpha^{(2)}_{(k-1)n+j}/2) C^{(x)}_{K\chi_j} R_{z\chi_j}(\alpha^{(3)}_{(k-1)n+j}/2) C^{(x)}_{K\chi_j} R_{z\chi_j}(\alpha^{(3)}_{(k-1)n+j}/2),
\end{aligned} \tag{35}$$

where

$$\begin{aligned}
&C_{K\chi_m\chi_{m+1}} = P_{K\chi_m} \otimes \sigma^{(x)}_{\chi_{m+1}} + (I_{K\chi_m} - P_{K\chi_m}) \otimes I_{\chi_{m+1}}, \\
&P_{K\chi_m} = |1\rangle_K |1\rangle_{\chi_m}\,{}_K\langle 1|_{\chi_m}\langle 1|, \\
&C^{(x)}_{K\chi_j} = |0\rangle_K\,{}_K\langle 0| \otimes \sigma^{(x)}_{\chi_j} + |1\rangle_K\,{}_K\langle 1| \otimes I_{\chi_j}.
\end{aligned} \tag{36}$$

It is simple to check that this operator acts as I_χ or U_χ on the basis states of the subsystem $\chi\cup K$ including, respectively, the state $|0\rangle_K$ or $|1\rangle_K$ of the subsystem K. The depth of this operator is $O(M\log N) \ge O(N^2)$.

C. Derivatives of loss function and iterations of optimization parameters

The input data for optimization algorithms include the derivatives of the loss functions with respect to the optimization parameters $\alpha = \{\alpha_k^{(j)}\}$ and $\beta = \{\beta_k^{(j)}\}$. Since these parameters in our algorithm are introduced through the exponent of Pauli matrices $\sigma^{(y)}$ and $\sigma^{(z)}$ in Eq.(35), such derivatives can be expressed in terms of the loss function with shifted values of parameters which can be calculated using the quantum algorithm described in Sec.II B. Then, the whole set of calculated loss functions can be supplied to the classical algorithm which calculates the subsequent values of the optimization parameters for the next iteration step of the optimization algorithm.

Let $\hat\gamma = \{\hat\gamma_1,\dots,\hat\gamma_{6nM}\}$ be the set of all parameters α and β: $\hat\gamma = \{\alpha,\beta\}$. Then [47]

$$\frac{\partial L(\hat\gamma)}{\partial\hat\gamma_k} = \frac{1}{2}\big(L(\hat\gamma^{(k,+)}) - L(\hat\gamma^{(k,-)})\big),\quad k=1,\dots,6nM, \tag{37}$$

where $\hat\gamma^{(k\pm)}$ means the set of parameters $\hat\gamma$, in which $\hat\gamma_k$ is replaced with $\hat\gamma_k \pm \pi/2$ i.e.,

$$\hat\gamma^{(k,\pm)} = \hat\gamma|_{\hat\gamma_k\to\hat\gamma_k\pm\pi/2},\quad k=1,\dots,6nM. \tag{38}$$

Thus, we need to calculate the loss functions $L(\hat\gamma, L(\hat\gamma^{(k,\pm)})$, $k=1,\dots,6nM$, i.e., $12nM+1$ values all together.

After calculating derivatives (37), we can calculate the new values of the parameters $\hat\gamma$:

$$\hat\gamma_k \to \hat\gamma_k - \delta \frac{\partial L(\hat\gamma)}{\partial \hat\gamma_k}, \quad k = 1, \dots, 6nM, \tag{39}$$

where δ is some small parameter, and calculate subsequent set of $12nM + 1$ loss functions for this new set of optimization parameters and so on. The iteration stops when the algorithm reaches the required accuracy of calculating either eigenvalues or loss function. The relation between these two accuracies will be shown in Sec.IV.

III. FEATURES OF ALGORITHM FOR CALCULATING EV

Formally, the algorithm for problem on eigenvalues of the matrix A can be deduced from the described algorithm putting $B = I$ in Eq.(4). However, in this case it is natural to use the triangularization formula [66]

$$Q^\dagger A Q = T, \tag{40}$$

$$\lambda_k = t_{kk}, \tag{41}$$

where Q is the unitary matrix and T is the upper-triangular matrix, λ_k and t_{kk} are, respectively, the eigenvalue of A and the diagonal element of T. For this case, the loss function (4) should be replaced with

$$L(\beta) = \sum_{j=1}^{N-1} \sum_{j=0}^{i-1} |\langle i|U^\dagger(\beta) A U(\beta)|j\rangle|^2 = \sum_{j=1}^{N-1} \sum_{j=0}^{i-1} |\langle i|T(\beta)|j\rangle|^2, \tag{42}$$

reducing twice the number of optimization parameters. Since there is no matrix B, we do not need the one-qubit subsystem L, the circuit in Fig.1 must be modified by removing the subsystem L and replacing $U(\alpha) \to U^*(\beta)$, where $*$ denotes the complex conjugate. Now the number of parameters in the list $\hat\gamma$ is $3nM$: $\hat\gamma = \{\hat\gamma_1, \dots, \hat\gamma_{3nM}\} = \{\beta^{(n)}_j : n = 1, 2, 3, j = 1, \dots nM\}$. Since all the steps of the algorithm with such simplified loss function are exactly the same, we do not give more details of the quantum algorithm for calculating the loss function. If the optimizing values β_0 of the β-parameters are found, then $Q = U(\beta_0)$.

However, the formula (37) for the derivatives of the loss function must be modified. Since each parameter from the list $\hat\gamma$ appears four times in the expression for the loss function (42), the formula for the derivative of the loss function with respect to the optimization parameters becomes as follows:

$$\frac{\partial L(\hat\gamma)}{\partial \hat\gamma_k} = -\frac{\sin^2 \alpha/2}{2\cos\alpha} \left(2\Big(L(\hat\gamma^{k,+}) - L(\hat\gamma^{k,-}) \Big) - \frac{1}{\sin^2 \alpha/2 \sin\alpha} \Big(L(\hat\gamma^{k,+\alpha}) - L(\hat\gamma^{k,-\alpha}) \Big) \right), \tag{43}$$

where $\hat\gamma^{(k,\pm)}$ are defined in (38), and $\hat\gamma^{(k,\pm\alpha)}$ denotes the set of parameters $\hat\gamma$, in which $\hat\gamma_k$ is replaced with $\hat\gamma_k \pm \alpha$, $\alpha \ge 0$, $\alpha \ne n\pi/2$, $n \in \mathbb{Z}$. Thus, we calculate the set of loss functions including L, $L(\hat\gamma^{k,\pm})$, $L(\hat\gamma^{k,\pm\alpha})$, $k = 1, \dots, 3nM$, i.e., $12nM + 1$ values all together, the same as in Sec.II. Then, we proceed to calculating the derivatives of the loss function by using Eq.(43), constructing subsequent values of the optimization parameters by using Eq.(39), calculate the subsequent set of $12nM + 1$ loss functions for the new set $\hat\gamma$ by using the quantum algorithm and so on. The iteration process stops when the required accuracy of calculation is achieved.

IV. CHARACTERISTICS OF ALGORITHMS

Both algorithms considered in Secs.II and III have similar parameters characterizing their effectiveness.

1.*The depth* of the circuit is defined by the operators $W^{(2)}_{\chi\psi K}$ and $W^{(5)}_{\tilde\chi\tilde\psi K\tilde B}$ whose depths are $O(M\log N)$ and $O(N^2 \log N)$ respectively. Depending on M, either of two depth can prevail.

2. *The space* required for realization of quantum algorithm is $O(\log N)$ qubits.

4. *The success probability* p, $p \sim N^{-4}$ according to Eq.(30), is the probability of removing the garbage via the measurement of the state of the ancilla B.

3. *The accuracy* ε *of calculating the eigenvalues* is the principal parameter characterizing the accuracy of calculation. It is related to other accuracies listed below.

4. *The number of iterations* $N^{(it)}$ needed to obtain the eigenvalues with required accuracy ε. The numerical simulations in Sec. V show that, at least for the considered example, $N^{(it)} = -\varkappa_0 \log C_0\varepsilon$, where the parameters $\varkappa_0$ and C_0 are defined by the matrices A and B used in generalized eigenvalue problem, by the parameter M in the unitary transformations U_χ, U_ψ and by particular gradient method implemented in the algorithm.

5. *The accuracy ε_L of calculating the loss function* characterizes the deviation of the optimized L from zero: $L \le \varepsilon_L$.

6. *The number of runs of the algorithm.* To probabilistically find the loss function $L(\hat\gamma)$ and all $L(\hat\gamma^{(k)})$ needed for calculating the derivatives at fixed values of the parameters, we have to perform $N_1^{(r)} = 12nM+1$ runs of the algorithm, as shown in Secs. II C, III. In addition, each iteration requires $N_2^{(r)} \sim \frac{1}{\varepsilon_p}$ runs to find probability p_1 in Eq.(33) with the accuracy ε_p ($p_1 = p_1^{(0)} + \varepsilon_p$). The total number of runs is $N^{(r)} = N_1^{(r)} N_2^{(r)}$. In turn, ε_p is related to the accuracy ε_L required for probabilistic calculation of the loss function L. This relation follows from Eq.(33):

$$L = a\frac{p_1}{p_0} = a\frac{p_1^{(0)}+\varepsilon_p}{1-p_1^{(0)}-\varepsilon_p} \approx \frac{ap_1^{(0)}}{1-p_1^{(0)}} + \Delta L, \tag{44}$$
$$a = \left(|\tilde a_{00}|^2 + |\tilde b_{00}|^2\right),\quad \Delta L = \frac{a}{(1-p_1^{(0)})^2}\varepsilon_p,$$

where we assume that $a \neq 0$. Obviously, to provide the accuracy ε_L, we require

$$\Delta L \lesssim \varepsilon_L. \tag{45}$$

8. *The accuracy σ of encoding the parameters $\hat\gamma$.* The optimization parameters $\hat\gamma$ in the unitary transformations U_χ and U_ψ can not be realize with absolute accuracy. We introduce the parameter $\sigma = 10^{-d}$, where d is the number of decimals kept in the parameters $\hat\gamma$, thus $\hat\gamma_k = \hat\gamma_k^{(0)} + \hat\gamma_k^{(\sigma)}\sigma$, where $\hat\gamma_k^{(0)}$ is the encoded parameter and $\hat\gamma_k^{(\sigma)}$ characterizes the neglected part of the parameter $\hat\gamma_k$.

The accuracies of calculating the eigenvalues and loss function, parameters ε and ε_L, are used in the classical part of the algorithm. These two parameters can be related to each other as follows. Let the matrix elements of T and S in the formula for the loss function, Eq. (4), be found with the accuracy $\tilde\sigma \ll 1$, i.e., $t_{ij} = t_{ij}^{(0)} + t_{ij}^{(\tilde\sigma)}\tilde\sigma$, $s_{ij} = s_{ij}^{0)} + s_{ij}^{(\tilde\sigma)}\tilde\sigma$, where $t_{ij}^{(0)}$ and $s_{ij}^{(0)}$ are some mean values of the appropriate matrix elements, while $t_{ij}^{(\tilde\sigma)}$ and $s_{ij}^{(\tilde\sigma)}$ characterize deviations from the mean values. The loss function L includes only vanishing elements of T and S for which the mean values are zero, therefore

$$L = L^{(\tilde\sigma)}\tilde\sigma^2 = \varepsilon_L, \tag{46}$$

where $L^{(\tilde\sigma)} > 0$ can be written in terms of $t_{ij}^{(\tilde\sigma)}$ and $s_{ij}^{(\tilde\sigma)}$ (we do not present exact formula). The accuracy ε for the eigenvalues can be expressed in terms of $\tilde\sigma$ following Eq.(3) (we consider the accuracy for GEV; the accuracy for EV can be obtained similarly using Eq.(41)):

$$\lambda_k = \lambda_k^{(0)} + \lambda_k^{(\tilde\sigma)}\tilde\sigma,\quad \lambda_k^{(0)} = \frac{t_{kk}^{(0)}}{s_{kk}^{(0)}},\quad \lambda_k^{(\tilde\sigma)} = \left(\frac{t_{kk}^{(\tilde\sigma)}}{s_{kk}^{(0)}} - \frac{t_{kk}^{(0)}s_{kk}^{(\tilde\sigma)}}{(s_{kk}^{(0)})^2}\right), \tag{47}$$

where $\lambda_k^{(0)}$ is the exact GEV; $\lambda_k^{(\tilde\sigma)} = t_{kk}^{(\tilde\sigma)}$ in the case of Eq.(41) for EV. We can introduce ε as follows:

$$\varepsilon = \lambda^{(\tilde\sigma)}\tilde\sigma,\quad \lambda^{(\tilde\sigma)} = \frac{\tilde\sigma}{N}\sum_{k=1}^{N}|\lambda_k^{(\tilde\sigma)}|. \tag{48}$$

Comparison of (46) and (48) yields

$$\varepsilon_L = \frac{L^{(\tilde\sigma)}}{(\lambda^{(\tilde\sigma)})^2}\varepsilon^2. \tag{49}$$

The factor at ε^2 in the right hand side of (49) depends on a particular matrix, while the quadratic relation between ε_L and ε is universal.

The accuracy σ of encoding the parameters $\hat\gamma$ yields the lower boundary $\varepsilon^{(min)}$ for the accuracy of calculation of GEV and EV. To estimate this lower boundary, we note that σ-accuracy for $\hat\gamma$ results in perturbation of the operators U_χ and U_ψ: $U_\chi(\alpha) = U_\chi(\alpha^{(0)}) + \sum_{j>0} U_\chi^{(j)}\sigma^j$ and similar for $U_\psi(\beta)$. Substituting these perturbed unitary transformations into the loss function (4) we obtain, up to the first order in σ,

$$L = L(\gamma^{(0)}) + L^{(\sigma)}\sigma, \tag{50}$$

we do not present the expression for $L^{(\sigma)}$. Therefore, the minimal values of ε_L is a linear function of σ: $\varepsilon_L^{(min)} = L^{(\sigma)}\sigma$. Then Eq.(49) yields:

$$\varepsilon^{(min)} = \lambda^{(\sigma)}\sqrt{\sigma},\quad \lambda^{(\sigma)} = \lambda^{(\tilde\sigma)}\sqrt{\frac{L^{(\sigma)}}{L^{(\tilde\sigma)}}}. \tag{51}$$

Relations (49)-(51) are confirmed by numerical simulations in Sec.V.

V. NUMERICAL SIMULATIONS

We consider the problems on both GEV and EV for the 4×4 matrices A and B encoded into the states of the five-qubit subsystem $R\cup C\cup L$, where R and C are two-qubit subsystems ($n=2$). We randomly generate 100 pairs of the matrices A and B under condition that $\min_i |\lambda_i(A,B)|\ge 0.1$. Initially, we chose the parameter $\delta=0.5$ in Eq.(39) for the iterated parameters $\hat\gamma$. Then this parameter dynamically increases with the decrease in the value of the loss function according to the following rule:

$$\begin{aligned}&\text{if}\;\; \frac{2(L^{(n-1)}-L^{(n)})}{L^{(n-1)}+L^{(n)}}<0.001,\;\;\text{then}\;\; \delta\to 1.0005\,\delta;\\ &\text{if}\;\; L^{(n)}>L^{(n-1)},\;\;\text{then}\;\; \delta\to\frac{\delta}{1.0005},\end{aligned} \tag{52}$$

where $L^{(n)}$ is the value of the loss function at the nth iteration step. We set $M=10$ in Eq.(35), $\alpha=\pi/3$ in (43) , and all optimization parameters $\hat\gamma$ equal to zero initially.

A. Unperturbed optimization parameters $\hat\gamma$

In this section, we assume that the optimization parameters $\hat\gamma$ at each iteration step can be perfectly encoded into the unitary transformations U_χ and U_ψ. The optimization algorithm stops iterating if $1/4\sum_{i=1}^4|\lambda_i-\lambda_i^{(0)}|<\varepsilon$, where $\lambda_i^{(0)}$ are the (generalized) eigenvalues calculated by the classical algorithm. We obtain the iteration number N^{it} and logarithm of the loss function $L=\varepsilon_L$, averaged over experiments with 100 pairs of random 4×4 matrices $\{A,B\}$, as functions of $\log\varepsilon$ for $\varepsilon=10^{-j}$, $j=1,\dots,7$. The averaged results are shown by the red dots on Fig. 2. In this section, log means the logarithm to the base 10. For the family of the above 100 numerical experiments and for small enough ε ($\varepsilon\le 10^{-3}$ in the considered example), $N^{(it)}$ can be approximated by the linear function

$$N^{(it)}=-\varkappa^{(N)}\log\varepsilon-C^{(N)}, \tag{53}$$

where $\varkappa_N$ and C_N can be written as follows:

$$\varkappa^{(N)}\approx\varkappa^{(N;0)}\pm\Delta\varkappa^{(N)},\;\; C^{(N)}\approx C^{(N;0)}\pm\Delta C^{(N)}. \tag{54}$$

Here the superscript 0 denotes the mean value over 100 experiments and Δ denotes the mean-square deviation of the appropriate parameter, for instance,

$$\varkappa^{(N;0)}=\frac{1}{100}\sum_{j=1}^{100}\varkappa_j^{(N)},\;\;\Delta\varkappa^{(N)}=\sqrt{\frac{1}{100}\sum_{j=1}^{100}(\varkappa_j^{(N)}-\varkappa^{(N;0)})^2}, \tag{55}$$

where the parameter $\varkappa_j^{(N)}$ is associated with the jth pair of the matrices $\{A,B\}$. In our example

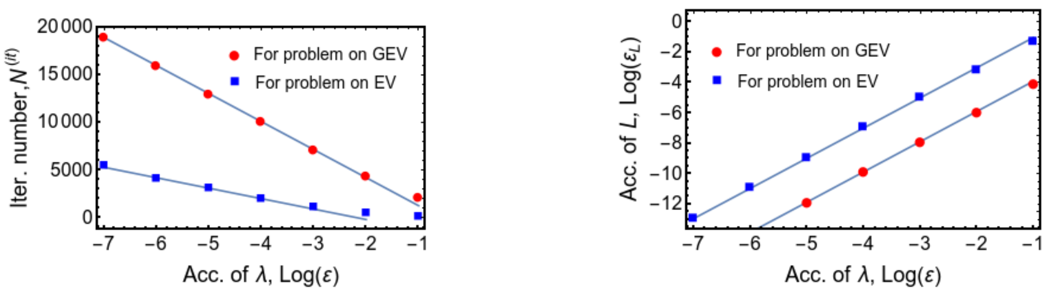
$$\begin{aligned}&\text{for problem on GEV:}\quad \varkappa^{(N;0)}=2936.66,\;\; C^{(N;0)}=1670.77,\;\;\Delta\varkappa^{(N)}=2229.85,\;\;\Delta C^{(N)}=4257.67,\\ &\text{for problem on EV:}\quad \varkappa^{(N;0)}=1094.51,\;\; C^{(N;0)}=2373.77,\;\;\Delta\varkappa^{(N)}=1882.98,\;\;\Delta C^{(N)}=5939.05.\end{aligned} \tag{56}$$

The functions $N^{(it;0)}=-\varkappa^{(N;0)}\log\varepsilon-C^{(N;0)}$ for the both algorithms are shown in Fig.2a by the lines. The red circles and blue squares on this graph well fit the appropriate lines for $\varepsilon\le 10^{-3}$. The large mean-square deviation of both parameters $\varkappa^{(N)}$ and $C^{(N)}$ means that the lines approximating dependence $N^{(it)}(\varepsilon)$ are completely different for different pairs of matrices $\{A,B\}$. In addition, the iteration number $N^{(it)}$ depends on the parameter M in the unitary transformation and decreases with an increase in M (till some saturation value), but the increase in M means the increase in the depth of the algorithm, therefore one has to manipulate between M and $N^{(it)}$ to get optimal evaluation time for the algorithm.

Similarly, the parameter $\log\varepsilon_L$ as a function of $\log\varepsilon$ for the considered family of 100 numerical experiments can be also approximated by the linear function for $\varepsilon\le 10^{-3}$:

$$\begin{aligned}&\log\varepsilon_L=\varkappa^{(L)}\log\varepsilon-C^{(L)}\;\;\Rightarrow\;\;\varepsilon_L=10^{-C^{(L)}}\varepsilon^{\varkappa^{(L)}},\\ &\varkappa^{(L)}=\varkappa^{(L;0)}\pm\Delta\varkappa^{(L)},\;\; C^{(L)}=C^{(L;0)}\pm\Delta C^{(L)},\\ \text{for problem on GEV:}\;\;&\varkappa^{(L;0)}=1.990,\;\; C^{(L;0)}=1.914,\;\;\Delta\varkappa^{(L)}=0.088,\;\;\Delta C^{(L)}=4.041,\\ \text{for problem on EV:}\;\;&\varkappa^{(L;0)}=1.986,\;\; C^{(L;0)}=0.953,\;\;\Delta\varkappa^{(L)}=0.071,\;\;\Delta C^{(L)}=1.203,\end{aligned} \tag{57}$$

where the superscript 0 and Δ mean the same as in Eqs. (54), see definitions in Eqs. (55). We emphasize that expression (57) for ε_L agrees with the quadratic dependence of ε_L on ε in Eq. (49) because $\varkappa^{(L)} \approx 2$ in this case; the mean-square deviation $\Delta\varkappa^{(L)}$ is $\approx 4\%$ of $\varkappa^{(L;0)}$ for both problems on GEV and on EV (i.e. the slope of all 100 lines in Eq.(57) is almost the same), unlike the mean-square deviation of the parameter $\varkappa^{(N)}$ in line (53) approximating $N^{(it)}$, see Eqs.(56). On the contrary, the mean-square deviation of $C^{(L)}$ is large (it exceeds the mean values of $C^{(L)}$), i.e., this parameter significantly depends on the considered matrices. We also emphasize that the slopes of lines for problems on GEV and EV on Fig.2b are very close to each other, which confirms Eq.(49) that holds for both algorithms.



(a) (b)

FIG. 2: (a) The iteration number N_r and (b) the accuracy of the loss function ε_L as functions of the accuracy $\log\varepsilon$ of constructed GEV and EV. These figures show that both N_r and $\log\varepsilon_L$ are proportional to $\log\varepsilon$ for small enough ε. The red circle and blue squares correspond, respectively, to the problem on GEV discussed in Sec.II, and to the problem on EV discussed in Sec.III. Slopes of both lines in plate (b) are very close to each other. The algorithm for constructing EV converges faster (iteration number is less)

B. Perturbation of optimization parameters $\hat{\gamma}$

As mentioned in Sec.IV, the optimization parameters $\hat{\gamma}$ can be implemented with the accuracy σ which determines the minimal accuracy for the eigenvalues $\varepsilon^{(min)}$. We emphasize that such approximate encoding of optimization parameters (which are parameters of one-qubit rotations) does not beak unitarity of U_χ and U_ψ. In this case, we break the optimization algorithm if the iterations stop converging the eigenvalues to their theoretically predicted values, i.e., the algorithm stops if the condition $1/4\sum_{i=1}^{4}|\lambda_i^{(n-1)} - \lambda_i^{(n)}| > 10^{-12}$ is destroyed. Here we use some conventional quantity 10^{-12} characterizing the convergency rate, $\lambda_i^{(n)}$ denotes the ith eigenvalue at the nth iteration step. For the same set of 100 pairs of matrices $\{A, B\}$ (or just of matrices A in the case of the problem on EV), we obtain the iteration number $N^{(it)}$, the logarithm of the loss function $L = \varepsilon_L$ and the logarithm of $\varepsilon^{(min)}$ averaged over 100 experiments as functions of $\log(\sigma)$. Results are shown by the red circles and blue squares for, respectively, the problems on GEV and EV on Fig.3.

Fig.3a demonstrates that the convergence rate is significantly larger in the case of algorithm for constructing EV. We emphasize that the slopes of the lines approximating the sets of red circles and blue squares in Fig.3b,c are wery close to each other that confirms Eq.(50) and Eq.(51) which hold for both algorithms.

In this case, $N^{(it)}$, $\log\varepsilon_L$ and $\log\varepsilon^{(min)}$ for the considered family of 100 experiments can be approximated by the linear

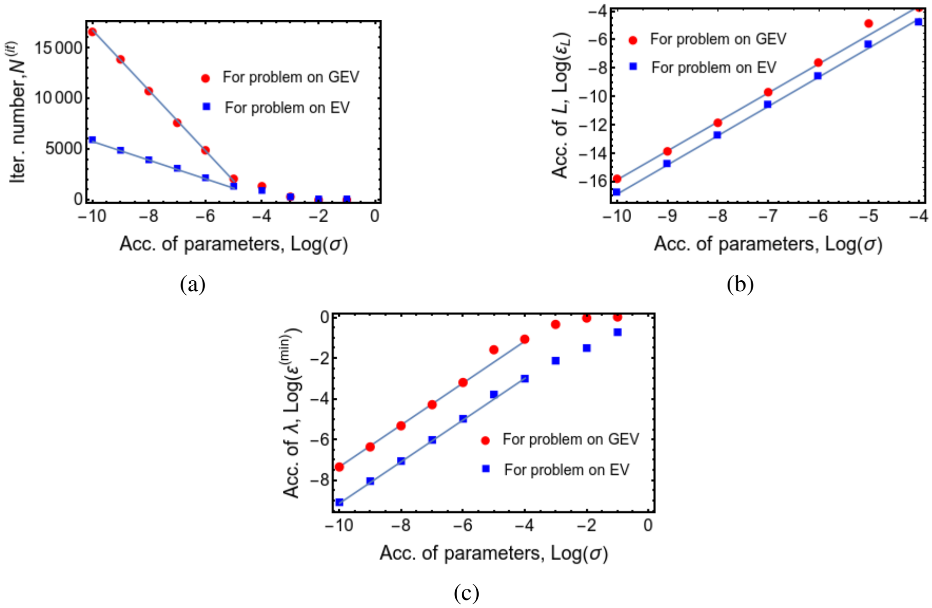


FIG. 3: (a) The iteration number N_r, (b) the accuracy of the loss function $\log \varepsilon_L$ and (c) the minimal reachable accuracy of eigenvalue $\log \varepsilon^{(min)}$ as functions of the accuracy of encoding the optimization parameters $\log \sigma$ for both problems on GEV and EV. These figures show that all presented quantities are proportional to $\log \sigma$ for small enough σ. The red circles and blue squares correspond, respectively, to the problem on GEV discussed in Sec.II, and to the problem on EV discussed in Sec.III. Slopes of both lines in plates (b) and (c) are very close to each other. Similar to Fig.2, the algorithm for constructing EV converges faster.

function of $\log \sigma$ for $\sigma \leq 10^{-6}$:

$$\begin{aligned}
& N^{(it)} = -\varkappa^{(N)} \log \sigma - C^{(N)}, \;\; \varkappa^{(N)} = \varkappa^{(N;0)} \pm \Delta\varkappa^{(N)}, \;\; C^{(N)} = C^{(N;0)} \pm \Delta C^{(N)}, \\
\text{for problem on GEV:} \quad & \varkappa^{(N;0)} = 2981.93, \;\; C^{(N;0)} = 13074.7, \;\; \Delta\varkappa^{(N)} = 2748.08, \;\; \Delta C^{(N)} = 18606.5, \\
\text{for problem on EV:} \quad & \varkappa^{(N;0)} = 922.76, \;\; C^{(N;0)} = 3456.64, \;\; \Delta\varkappa^{(N)} = 1395.32, \;\; \Delta C^{(N)} = 8746.82,
\end{aligned} \tag{58}$$

$$\begin{aligned}
& \log \varepsilon_L = \varkappa^{(L)} \log \sigma + C^{(L)} \;\; \Rightarrow \;\; \varepsilon_L = 10^{-C^{(L)}} \sigma^{\varkappa^{(L)}} \\
& \varkappa^{(L)} = \varkappa^{(L;0)} \pm \Delta\varkappa^{(L)}, \;\; C^{(L)} = C^{(L;0)} \pm \Delta C^{(L)}, \\
\text{for problem on GEV:} \quad & \varkappa^{(L;0)} = 2.034, \;\; C^{(L;0)} = 4.528, \;\; \Delta\varkappa^{(L)} = 0.245, \;\; \Delta C^{(L)} = 3.509, \\
\text{for problem on EV:} \quad & \varkappa^{(L;0)} = 2.053, \;\; C^{(L;0)} = 3.714, \;\; \Delta\varkappa^{(L)} = 0.240, \;\; \Delta C^{(L)} = 2.919,
\end{aligned} \tag{59}$$

$$\begin{aligned}
& \log \varepsilon^{(min)} = \varkappa^{(\lambda)} \log \sigma + C^{(\lambda)} \;\; \Rightarrow \;\; \varepsilon = 10^{-C^{(\lambda)}} \sigma^{\varkappa^{(\lambda)}}, \\
& \varkappa^{(\lambda)} = \varkappa^{(\lambda;0)} \pm \Delta\varkappa^{(\lambda)}, \;\; C^{(\lambda)} = C^{(\lambda;0)} \pm \Delta C^{(\lambda)}, \\
\text{for problem on GEV:} \quad & \varkappa^{(\lambda;0)} = 1.028, \;\; C^{(\lambda;0)} = 2.976, \;\; \Delta\varkappa^{(\lambda)} = 0.116, \;\; \Delta C^{(\lambda)} = 2.308, \\
\text{for problem on EV:} \quad & \varkappa^{(\lambda;0)} = 1.024, \;\; C^{(\lambda;0)} = 1.131, \;\; \Delta\varkappa^{(\lambda)} = 0.114, \;\; \Delta C^{(\lambda)} = 1.145.
\end{aligned} \tag{60}$$

Similar to the $N^{(it)}$ in Sec.V A, the parameters $\varkappa^{(N)}$ and $C^{(N)}$ of linear approximation for $N^{(it)}$ in Eq.(58) have large mean-square deviations, so that the approximating lines are completely different for different pairs of matrices $\{A, B\}$ (or for different matrices A in the case of problem on EV). The mean-square deviation of the parameters $\varkappa^{(L)}$ and $\varkappa^{(\lambda)}$ (slopes of the lines) in, respectively, Eqs.(59) and (60) are relatively small, they take, respectively, $\approx 12\%$ and $\approx 11\%$ of the appropriate mean values for both problems on GEV and EV. This agrees with formulae (50) and (51) for ε_L and $\varepsilon^{(min)}$, because $\varkappa^{(L)} \approx 2$ and $\varkappa^{(\lambda)} \approx 1$ in formulae, respectively, (59) and (60). The mean-square deviations of $C^{(L)}$ and $C^{(\lambda)}$ are large, similar to mean-square deviation for $C^{(L)}$ in (57), i.e., these parameters significantly depend on the considered matrices A and B.

VI. CONCLUSIONS

We apply the method of matrix encoding into the quantum state to the quantum algorithm of calculating the loss function in frames of the variational algorithm for constructing of both GEV and EV. The matrices A and B under consideration are encoded into the superposition state of the subsystems R, C and L, at that the state of the one-qubit subsystem L labels the parts of the superposition state associated with the matrices A and B. This algorithm is applicable to arbitrary pair of complex square matrices A and B. However, in this paper, we consider the non-degenerate matrices and use the lower boundary 0.1 for the numerical simulation of GEV and EV. We remove the garbage acquired during calculations via measurement of the state of the ancilla B with the probability of desired output (success probability) $\sim N^{-4}$. The depth of the circuit is estimated by either $O(M \log N)$ or $O(N^2 \log N)$ depending on M, while the space is $O(\log N)$. We emphasize that the estimation $O(N^2 \log N)$ is related to selecting the vanishing elements from the matrices T and S in expression (4) for the loss function.

After removing the garbage, the loss function can be obtained probabilistically performing the set of $\sim 1/\varepsilon_L$ runs so that the loss function becomes expressed in terms of the probability amplitudes of the superposition state which originally contained the elements of the matrices A and B, see Eqs.(31) - (33). The number of runs needed to calculate the loss function together with all necessary derivatives with respect to the optimization parameters is $\sim M \log N/\varepsilon_L$, where $M \log N$ is the estimated number of optimization parameters and the accuracy ε_L of L depends on the desired accuracy ε of GEV (or EV). We show that $\varepsilon_L \sim \varepsilon^2$. The accuracy σ of encoding the optimization parameters imposes the lower boundary $\varepsilon^{(min)}$ on the accuracy of the calculated eigenvalues and $\varepsilon^{(min)} \sim \sqrt{\varepsilon_L} \sim \sqrt{\sigma}$. The above theoretically obtained relations among accuracy-parameters ε, ε_L and σ are confirmed by numerical simulations with 100 pairs of square random complex 4×4 matrices A and B for the problem on GEV and with 100 matrices A for the problem on EV. The advantage of matrix-encoding method is that it is applicable to any matrices (subjected to normalization (7)) and does not require representing the input matrices as linear combinations of unitary matrices or embedding them into the Hermitian matrix with subsequent exponentiating to obtain a unitary operator.

One also has to remember the success probability on the step of garbage removing which is $O(N^{-4})$ and creates certain problem. But our expectation is that such problem having quantum nature can be overcome via quantum methods. A possible variant might be the so-called controlled measurement [69], which would allow to resolve the problem of small success probability with minimal expenses. However, this method still requires justification.

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